MILITARY SPECIFICATION

NITROCELLULOSE PAPER

1. SCOPE

1.1 This specification covers three types of paper designated as Nitrocellulose paper.

1.2 Classification.-Nitrocellulose paper shall be classified according to the following nominal percentages of nitrocellulose and cellulose, woodpulp (sulphate) on the total fiber basis:

Type:	I	II	III
Nitrocellulose, percent	50 ± 2.0	75 ± 2.0	85 ± 2.0
Cellulose, Woodpulp (Sulphate, Percent)	50 ± 2.0	25 ± 2.0	15 ± 2.0

2. APPLICABLE DRAWINGS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-D-98 - Diphenylamine.
MIL-A-140 - Adhesive, Water-Resistant, Waterproof Barrier Material.
MIL-N-244 - Nitrocellulose, (for use in ammunition).
MIL-C-20330 - Cellulose, Woodpulp (Sulphate).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-286 - Propellants: Sampling, Inspection and Testing.
MIL-STD-1234 - Pyrotechnics: Sampling, Inspection and Testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

PUBLICATIONS

ORDNANCE

ORD M608-11 - Procedures and Tables for Continuous Sampling by Attributes.

(Copies of Specifications, Standards, Drawings and Publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publications should cite "49 CFR 71-90 (latest revision)".

3. REQUIREMENTS

3.1 Materials.-The constituent materials shall comply with the requirements of the applicable specifications as follows:

Constituent Material	Conforming to Specification
Nitrocellulose	MIL-N-244 Grade A, Type II
Cellulose, Woodpulp, (sulphate)	MIL-C-20330
Methylcellulose	Commercial Grade (see 6.3)
Diphenylamine	MIL-D-98

3.2 Heat Test at 134.5 degrees Centigrade (°C).-The nitrocellulose shall not change the color of the methyl violet test paper to a salmon-pink color in less than 30 minutes as specified in 4.3.1.

3.3 Additives.-Each type of nitrocellulose paper shall have additives based on the total fiber content as follows:

Methylcellulose, Percent:	1.5 ≤ 0.2
Diphenylamine, Percent:	1.0 ≤ 0.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

3.4 Moisture.-Moisture content of the finished paper shall be not less than 25 percent nor more than 35 percent by weight. The test shall be performed as specified in 4.3.2.5.

3.5 Form and dimensions.-The paper shall be furnished in rolls of the width and length as specified in the contract or order (see 6.1).

3.6 Workmanship.-

3.6.1 Rolls before wrapping.-The rolls of nitrocellulose paper shall be free from dirt, grit, water, oil, grease and other foreign material.

3.6.2 Rolls after wrapping.-The rolls of nitrocellulose paper and wrapping shall be free from water, oil, grease and other foreign material that may be absorbed by the wrapping.

3.7 Thickness.-The thickness of the nitrocellulose paper shall be as specified in the contract and shall be controlled to $\leq$ 0.002 inches (see 4.2.2.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 General Quality Assurance Provisions.-The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure suppliers and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system.-If the contractor desires to utilize a quality assurance system, which is at variance with the quality assurance provisions of 4.2 and 4.3 and other documents referenced herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspections, (measuring and testing equipment), and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance provisions required by the provisions of 4.2 and 4.3 and the other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

to whether certain procedures of the contractor's system provide equal assurance, the comparable procedures of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any changes to the written procedure that effects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
b. Quantity of product inspected.
c. Results obtained for all inspection performed.
d. Specification number and date, together with an identification and date of changes.
e. Certificates of analysis on all material procured directly by the contractor, when such material is covered by referenced Government specification.
f. Quantity of product in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure (see 4.1.1), this detail specification, and other contractual documents as a guide, the Government inspector shall verify all quality assurance operations performed by the contractor. Verification shall be in accordance with a. or b. as applicable, the decision being the responsibility of the procuring activity. In either case, the inspector shall also ascertain, prior to acceptance, that all quality assurance provisions of other specifications referenced in any of the contractual documents have been complied with. Deviations from prescribed or agreed upon procedures discovered by the Government inspector shall be brought to the attention of the supplier. Disposition of the product and remedial action shall be as directed by the Government inspector

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

and, depending on the nature of the deviation, may consist of lot rejection, screening, re-sampling, re-instruction of the supplier's employees, or other appropriate action:

a. Verification at the point of manufacture shall be accomplished at unscheduled intervals in accordance with 4.1.3.1 and 4.1.3.2.

b. Verification at the point of delivery shall be in accordance with 4.1.3.2.

4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or of approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection.-Product inspection shall consist of Government inspection of product which has been previously inspected by the contractor and found to meet the quality assurance provisions of this specification. The inspection by the Government shall be performed in order to determine that the product is of the quality required by this specification and that the contractor's records are reliable.

4.2 Inspection provisions.-

4.2.1 Lot formation.-A lot shall consist of one or more batches of Nitrocellulose Paper, produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of Nitrocellulose Paper that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

4.2.2 Examination.-Sampling plans and procedures for the following classification of defects shall be in accordance with Standard MIL-STD-105. Continuous sampling plans, in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity. AQL's and sampling plans may be applied to the individual characteristics listed using and AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

4.2.2.1 Rolls, prior to winding.

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Width	Gage	01001
102.	Thickness (see 3.7)	Gage	01002
Minor:	AQL 1.0 percent		
201.	Length, min	Gage	01003
202.	Evidence of poor workmanship (see 3.6)	Visual	01004

4.2.2.2 Rolls, after wrapping and sealing.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 2.50 percent		
201.	Protective wrapping torn, cut, perforated, or incomplete	Visual	02001
202.	Adhesive missing, incomplete or improperly positioned	Visual	02002
203.	Tape incorrect type, missing, incomplete or improperly positioned	Visual	02003
204.	Marking incorrect, misleading or unidentifiable	Visual	02004
205.	Evidence of poor workmanship (see 3.6)	Visual	02005

4.2.3 Testing

4.2.3.1 Sampling for test.-A random sample equal to 10 percent of the lot size shall be selected and a sample specimen 2 yards in length full width of the roll shall be taken from each sample unit for test purposes. The sample specimen shall be cut in strips, one inch in width, to fill an airtight one quart container for testing as specified in 4.3. The container shall be labeled to show the designation, name of manufacturer, contract or purchase order number, number of pounds in the lot, and lot number. If the sample fails to comply with any of the requirements, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

4.3 Test methods and procedures

4.3.1 Heat test at 134.5°C.-The heat test shall be performed in accordance with standard MIL-STD-286, method 404.1. The nitrocellulose shall have been tested not more than 30 days prior to its use. Code No. 03001.

4.3.2 Analysis of paper.-The paper shall be analyzed for nitrocellulose, woodpulp, diphenylamine, methylcellulose and moisture as described in 4.3.2.1 through 4.3.2.5. Code No. 04001

4.3.2.1 Nitrocellulose.-A sample containing approximately 250 milligrams of nitrocellulose shall be placed in a 300 milliliters titration flask fitted with a side-arm for the admission of carbon dioxide. Twenty-five milliliters of n-butyl acetate and forty-five milliliters of glacial acetic acid shall be added. The flask shall be attached to a source of carbon dioxide and refluxed for 5 minutes. Ten milliliters of concentrated hydrochloric acid and 25 milliliters of 0.7 Normal ferrous ammonium sulfate (MIL-STD-286)shall be added through the condenser and refluxing continued for 20 minutes. The flow of carbon dioxide shall be increased slightly and the flask cooled in a water-bath. Five milliliters of 20 percent ammonium thiocyanate shall be added and the solution titrated to the disappearance of the red color using 0.2 Normal titanous chloride (MIL-STD-286). Code No. 05001

A blank shall be run on the reagents.

$$\text{Nitrocellulose, percent} = \frac{46.69\ (V-B)\ N}{PW}$$

Where: P = Percent nitrogen in the nitrocellulose.
V = Volume of titanous chloride used for the sample.
B = Volume of titanous chloride used for the blank.
W = Weight of sample on a moisture free basis.
N = Normality of titanous chloride solution.

4.3.2.2 Woodpulp.-An accurately weighed gram sample shall be extracted in a tared extra-coarse tall-form sintered glass filtering crucible using a Reweg condenser and ethylene chloride (symmetrical dichloro ethane) as the solvent. The extraction shall be performed on an electric hot-plate for a period of 24 hours, after which the thimble shall be aspirated for a few minutes and dried to constant weight at 100°C. The extraction apparatus shall be reassembled, the extraction repeated for 24 hours, and the thimble again aspirated and dried to constant weight. A loss of less than 3 milligrams between extractions shall be considered evidence of complete extraction. The increase in weight of the thimble shall be calculated as percent nitrocellulose plus woodpulp. The percent woodpulp shall be obtained by subtracting the percent nitrocellulose determined from this value. The ethylene

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

chloride extract shall be blown down and saved for the dyphenylamine determination. Code No. 06001.

4.3.2.3 Diphenylamine.- The dried extract from the determination of wood-pulp shall be used and diphenylamine shall be determined by bromination as described in Standard MIL-STD-286, 201.1. Code No. 07001.

4.3.2.4 Methylcellulose.-The methylcellulose content shall be determined by adding the percentages of the other ingredients and subtracting from 100. Code No. 08001.

4.3.2.5 Moisture.-The moisture shall be determined by the Karl Fischer Method as specified in Standard MIL-STD-1234, Method 101.2. Code No. 09001.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level C.-Nitrocellulose paper shall be uniformly wet with a minimum of twenty-five pounds of water to seventy-five pounds of dry material. Unless otherwise specified, rolls, supported on cores, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipments shall comply with Interstate Commerce Commission Regulations 49 CFR 71-90, covering rules and regulations for transportation of explosives and other dangerous articles.

5.2 Marking.-Unless otherwise specified, markings shall be in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification.

6.2 Inspection code number.-The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.3 The methylcellulose used shall be similar to or as provided by Dow Corning Chemical Company, Midland, Michigan. A standard Grade MC, with a viscosity range of 13-18 centipoises. The viscosity is determined at 20 degrees Centigrade by dissolving the methylcellulose in water to a concentration of 2 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, The United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
ARMY-MU

Preparing Activity:
ARMY-MU

Project No. 1375-A506

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.